

# CS 486/686

# Hidden Markov Models I

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Lecture 10

RN 14.1 & 14.2.1 · PM 8.5.1–8.5.3

Search ›

Uncertainty ›

Decisions ›

Learning

## Learning goals

- **Construct** a hidden Markov model from a real-world scenario.
- State the **independence assumptions** baked into an HMM.
- Compute the **filtering probability** for a time step.
- **Justify** each step in the derivation of the filtering formula.

# Inference in a changing world

So far, we reasoned about a *static* world. But the world evolves over time — we need to reason about **sequences of events**.



Weather



Stocks



Patient monitor



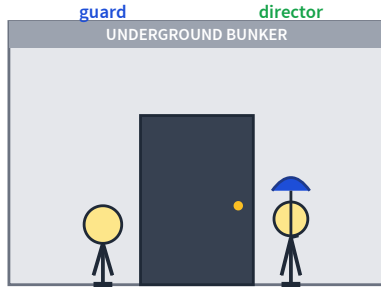
Robot  
localization



Speech

Today's running example: an **umbrella story** — the simplest possible HMM.

# The umbrella story



You are a **security guard** stationed at a secret underground facility.

You want to know whether it is **raining outside** today.

Your only signal: each morning, the director walks in **with** or **without** an umbrella.

# States and observations

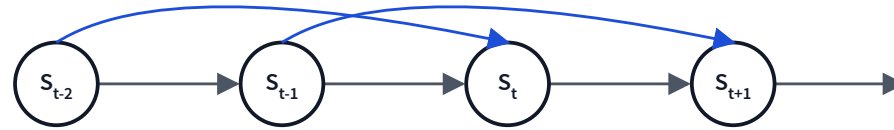
- The world contains a series of **time slices**.
- Each slice has a hidden **state**  $S_t$  and a visible **observation**  $O_t$ .

**Q.** What are the random variables in the umbrella world?

- $S_t = \text{true}$  iff it rains at time  $t$  ( hidden ).
- $O_t = \text{true}$  iff the director carries an umbrella at time  $t$  ( observed ).

# The Markov assumption

*"The future is independent of the past given the present."*



**First-order Markov** (the default):

$$P(S_t | S_{t-1}, \dots, S_0) = P(S_t | S_{t-1})$$

**Second-order:** also depends on  $S_{t-2}$  (blue skip-edges).

$$P(S_t | S_{t-1}, S_{t-2})$$

**Stationary process:** the same CPT is used at every time step.

# Transition model



## Transition CPT

$$P(s_0) = 0.5$$

$$P(s_t | s_{t-1}) = 0.7$$

$$P(s_t | \neg s_{t-1}) = 0.3$$

**Warm-up:** what is  $P(S_K)$  for large  $K$ ?

$$P(S_1 = T) = 0.5 \cdot 0.7 + 0.5 \cdot 0.3 = 0.5$$

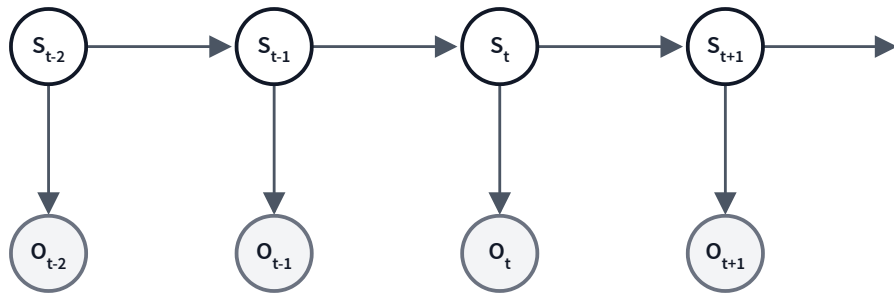
$P(S_K = T) = 0.5$  for all  $K$  – the chain reaches its steady state immediately.

# Sensor (observation) model

## Sensor Markov assumption

Each state is sufficient to generate its own observation:

$$P(O_t | S_t, S_{t-1}, \dots, O_{t-1}, \dots) = P(O_t | S_t).$$



## Sensor CPT

$$P(o_t | s_t) = 0.9$$

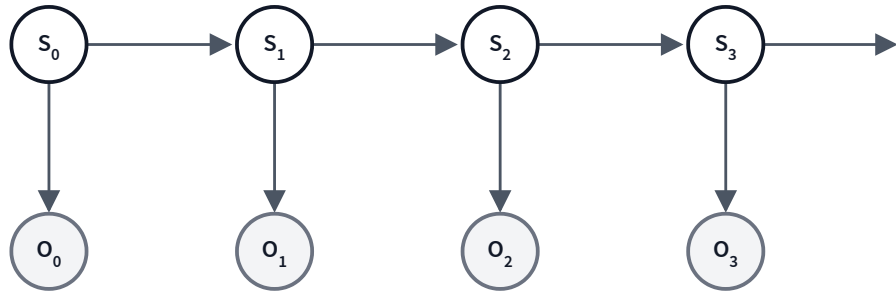
$$P(o_t | \neg s_t) = 0.2$$

**Convention:** in a CPT a lowercase letter means that variable is *true* ( $\neg$  = false) — e.g.  $P(o_t | s_t)$  is  $P(\text{umbrella} | \text{rain})$ .



# Complete HMM for the umbrella story

$S_t = T$  iff it rains on day  $t$ .  $O_t = T$  iff the director carries an umbrella on day  $t$ .



## Prior & transition

$$P(s_0) = 0.5$$

$$P(s_t | s_{t-1}) = 0.7$$

$$P(s_t | \neg s_{t-1}) = 0.3$$

## Sensor

$$P(o_t | s_t) = 0.9$$

$$P(o_t | \neg s_t) = 0.2$$

By the same reasoning as before,  $P(O_K = T) = 0.5 \cdot 0.9 + 0.5 \cdot 0.2 = 0.55$  for all  $K$  (steady state, no evidence).

# Hidden Markov Model

## HMM

- A **Markov process** over time (state  $S_t$  depends only on  $S_{t-1}$ ).
- States are **hidden**; we never observe them directly.
- Evidence  $O_t$  is **observable** and depends only on  $S_t$ .

An HMM is a Bayes net — so the **variable elimination algorithm** already works! But two specialized algorithms exploit the chain structure:

- **Forward-backward** — filtering & smoothing
- **Viterbi** — most-likely explanation

# Four common inference tasks

## ✳ Filtering — *now*

What state am I in **right now**, given all evidence to date?

$$P(S_k | o_{0:k})$$

## → Prediction — *tomorrow*

What state will I be in in the **future**?

$$P(S_{k+j} | o_{0:k}), \quad j > 0$$

## ↶ Smoothing — *yesterday*

What state was I in at some **past** time?

$$P(S_j | o_{0:k}), \quad j < k$$

## ✕ Most-likely explanation

Which **sequence** of states is most likely?

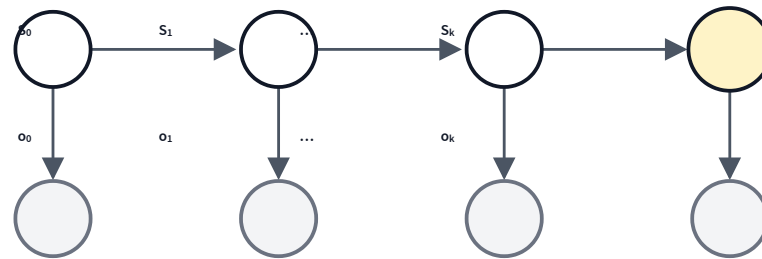
$$\arg \max_{s_{0:k}} P(s_{0:k} | o_{0:k})$$

Today: **filtering**.

# Filtering

Given observations from time 0 to time  $k$ , what is my distribution over the state *now*?

$$P(S_k | o_{0:k})$$



Yellow = query, gray = observed.  $o_{0:k}$  denotes the **observed values** we condition on (the evidence).

# From conditional to joint

By the definition of conditional probability:

$$P(S_k \mid o_{0:k}) = \frac{P(S_k, o_{0:k})}{P(o_{0:k})}$$

The denominator  $P(o_{0:k})$  is the same for every value of  $S_k$  — a normalizing constant  $\alpha$ :

$$P(S_k \mid o_{0:k}) \propto P(S_k, o_{0:k})$$

Recover that joint by summing out the hidden states  $s_0, \dots, s_{k-1}$ :

$$P(S_k, o_{0:k}) = \sum_{s_0} \cdots \sum_{s_{k-1}} P(s_0, \dots, s_{k-1}, S_k, o_{0:k})$$

# Naive: enumerate the joint

Factor the joint with the HMM structure ( $s_k = S_k$ ):

$$P(S_k | o_{0:k}) \propto \sum_{s_0} \cdots \sum_{s_{k-1}} P(s_0) P(o_0 | s_0) \prod_{i=1}^k P(s_i | s_{i-1}) P(o_i | s_i)$$

**Tiny case** ( $k = 1$ ):

$$P(S_1 | o_0, o_1) \propto \sum_{s_0} P(s_0) P(o_0 | s_0) P(S_1 | s_0) P(o_1 | S_1)$$

3 mul  $\times$  2 + 1 add, all times 2  $\Rightarrow$  **14 ops** — manageable.

## ... but it blows up

For general  $k$ , the sum ranges over *all*  $2^k$  hidden-state assignments.

### Naive enumeration

Sum over all  $2^k$  hidden state assignments.

$O(k \cdot 2^k)$  **ops**

### Forward recursion (today)

Reuse  $f_{0:(k-1)}$  when computing  $f_{0:k}$ .

$O(k)$  **ops**

# Filtering by forward recursion

$$\text{Let } f_{0:k} \triangleq P(S_k | o_{0:k}).$$

**Base case ( $k = 0$ )**

$$f_{0:0} = \alpha P(o_0 | S_0) P(S_0)$$

**Recursive case ( $k \geq 1$ )**

$$f_{0:k} = \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k | s_{k-1}) f_{0:(k-1)}$$

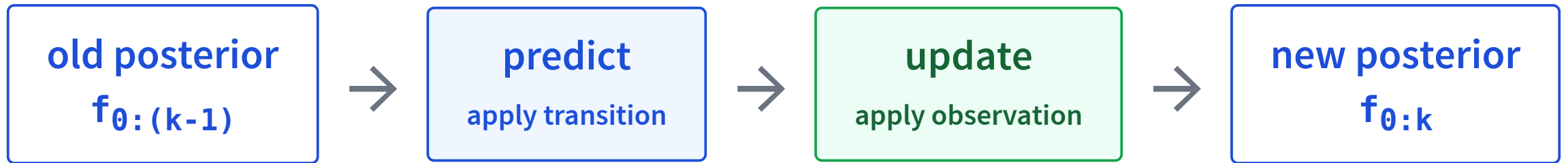
Total cost:  $O(k)$  ops — exponentially faster than naive enumeration.

$\alpha$  is a normalization constant (chosen so the two entries of  $f_{0:k}$  sum to 1).



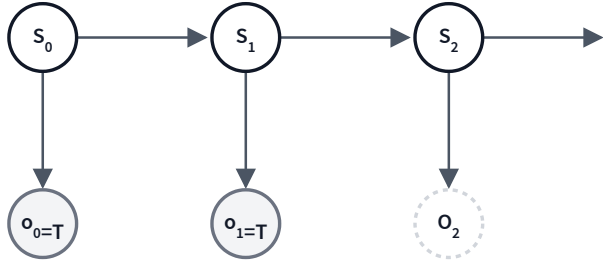
# Filtering = predict + update

$$f_{0:k} = \underbrace{\alpha P(o_k | S_k)}_{\text{update}} \cdot \underbrace{\sum_{s_{k-1}} P(S_k | s_{k-1}) f_{0:(k-1)}}_{\text{predict}}$$



Recursive Bayesian update — one step at a time.

# A filtering example: setup



The umbrella story. Given  $O_0 = T$ ,  $O_1 = T$  (umbrella seen on both days), compute  $f_{0:0}$  and  $f_{0:1}$  using forward recursion.

## Numbers

$$P(s_0) = 0.5$$

$$P(s_t | s_{t-1}) = 0.7, \quad P(s_t | \neg s_{t-1}) = 0.3$$

$$P(o_t | s_t) = 0.9, \quad P(o_t | \neg s_t) = 0.2$$

**Base case:**  $f_{0:0} = P(S_0|o_0)$

Use  $f_{0:0} = \alpha P(o_0|S_0) P(S_0)$ , treating each as a 2-vector  $\langle T, F \rangle$ .

$$f_{0:0} = \alpha \langle P(o_0|s_0), P(o_0|\neg s_0) \rangle \cdot \langle P(s_0), P(\neg s_0) \rangle$$

$$= \alpha \langle 0.9, 0.2 \rangle \cdot \langle 0.5, 0.5 \rangle$$

$$= \alpha \langle 0.45, 0.1 \rangle$$

$$= \langle 0.818, 0.182 \rangle$$

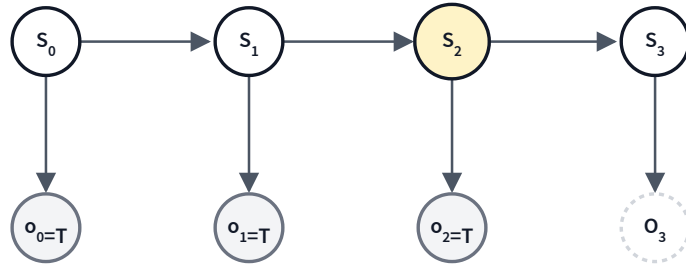
Normalize:  $0.45/(0.45 + 0.1) \approx 0.818$ ;  $0.1/0.55 \approx 0.182$ .

**Recursive case:**  $f_{0:1} = P(S_1 | o_0, o_1)$

$$\begin{aligned} f_{0:1} &= \alpha P(o_1 | S_1) \sum_{s_0} P(S_1 | s_0) f_{0:0} \text{ with } f_{0:0} = \langle 0.818, 0.182 \rangle. \\ &= \alpha \langle 0.9, 0.2 \rangle \cdot (\langle 0.7, 0.3 \rangle \cdot 0.818 + \langle 0.3, 0.7 \rangle \cdot 0.182) \\ &= \alpha \langle 0.9, 0.2 \rangle \cdot (\langle 0.5726, 0.2454 \rangle + \langle 0.0546, 0.1274 \rangle) \\ &= \alpha \langle 0.9, 0.2 \rangle \cdot \langle 0.6272, 0.3728 \rangle \\ &= \alpha \langle 0.56448, 0.07456 \rangle \\ &= \langle 0.883, 0.117 \rangle \end{aligned}$$

After two days of umbrellas, our belief that it rained today went from  $0.5 \rightarrow 0.818 \rightarrow 0.883$ .

# Try it: 4-step HMM — compute $P(S_2|o_0, o_1, o_2)$



## Numbers

$$P(s_0) = 0.4$$

$$P(s_t|s_{t-1}) = 0.7$$

$$P(s_t|\neg s_{t-1}) = 0.2$$

$$P(o_t|s_t) = 0.9$$

$$P(o_t|\neg s_t) = 0.2$$

$$f_{0:0} = \langle 0.75, 0.25 \rangle \rightarrow f_{0:1} = \langle 0.859, 0.141 \rangle \rightarrow \boxed{f_{0:2} \approx \langle 0.884, 0.116 \rangle}$$

# Deriving the filtering recursion

Six small steps from  $P(S_k | o_{0:k})$  to the forward-recursion formula:

$$P(S_k | o_{0:k})$$

$$= P(S_k | o_k, o_{0:(k-1)})$$

*rewrite*

$$= \alpha P(o_k | S_k, o_{0:(k-1)}) P(S_k | o_{0:(k-1)})$$

*Bayes' rule*

$$= \alpha P(o_k | S_k) P(S_k | o_{0:(k-1)})$$

*Markov (sensor)*

$$= \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k, s_{k-1} | o_{0:(k-1)})$$

*sum rule*

$$= \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k | s_{k-1}, o_{0:(k-1)}) P(s_{k-1} | o_{0:(k-1)})$$

*chain rule*

$$= \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k | s_{k-1}) f_{0:(k-1)}$$

*Markov (transition)*

# Justify each step (1–3)

Pick one of: (A) rewrite (B) Bayes' rule (C) chain rule (D) Markov (E) sum rule

Q1.  $P(S_k | o_{0:k}) = P(S_k | o_k, o_{0:(k-1)})$

A — rewrite

Q2.  $= \alpha P(o_k | S_k, o_{0:(k-1)}) P(S_k | o_{0:(k-1)})$

B — Bayes' rule

Q3.  $= \alpha P(o_k | S_k) P(S_k | o_{0:(k-1)})$

D — Markov

# Justify each step (4–6)

Pick one of: (A) rewrite (B) Bayes' rule (C) chain rule (D) Markov (E) sum rule

Q4.  $\dots = \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k, s_{k-1} | o_{0:(k-1)})$

E — sum rule

Q5.  $= \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k | s_{k-1}, o_{0:(k-1)}) P(s_{k-1} | o_{0:(k-1)})$

C — chain rule

Q6.  $= \alpha P(o_k | S_k) \sum_{s_{k-1}} P(S_k | s_{k-1}) f_{0:(k-1)}$

D — Markov



# The big picture

$$f_{0:k} = \underbrace{\alpha P(o_k | S_k)}_{\text{update}} \cdot \underbrace{\sum_{s_{k-1}} P(S_k | s_{k-1}) f_{0:(k-1)}}_{\text{predict}}$$

- **Predict:** roll forward the previous posterior through the transition model.
- **Update:** multiply by the new observation likelihood and renormalize.
- Each step is  $O(1)$  in  $k$  (constant-size factors). Total:  $O(k)$ .

## Learning goals (recap)

- ✓ **Construct** a hidden Markov model from a real-world scenario.
- ✓ State the **independence assumptions** baked into an HMM.
- ✓ Compute the **filtering probability** for a time step.
- ✓ **Justify** each step in the derivation of the filtering formula.

## Next: HMMs, Part 2

- **Prediction** — what state will I be in tomorrow?
- **Smoothing** — what was my state yesterday, knowing today's evidence?
- **Most-likely explanation** — the *Viterbi* algorithm.

All three are variations on the same predict-update recursion.